1. **Item 6: Develop mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them**
   The question provides two plan diagrams. Students must read and extract the necessary dimensions (such as the garage's length, width, and roof height) from these diagrams. Correctly extracting this information is fundamental to solving the problem, making this the most critical aspect.
2. **Item 5: Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms**
   Calculating the roof area requires applying geometric rules such as the formula for the area of a rectangle and the Pythagorean theorem. These are familiar algorithms for the student. Only by applying these rules can the dimensions be converted into an area.
3. **Item 12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**
   The solution process involves multiple steps: first, calculating the slant length using the width and height; second, computing the area of one rectangular section; and finally, multiplying by 2 to get the total area. This demonstrates multi-step reasoning and procedural thinking.
4. **Item 8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**
   The question explicitly requires showing the work. This means the calculation process, including intermediate results, must be written out. This directly corresponds to articulating the solution and presenting the steps taken.
5. **Item 7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations**
   The calculations involve numerical operations (addition, subtraction, multiplication, division, square roots), algebraic expressions (the formula for the Pythagorean theorem), and the conversion of geometric information. Students need to be proficient in manipulating these mathematical objects.

6. **Item 10: Make sense of, relate and use a variety of representations when interacting with a problem**
   The plan diagrams are a form of graphical representation. Students must understand their relationship to the actual dimensions of the garage and translate the graphical information into numerical values. This demonstrates an understanding and use of different representations.
7. **Item 1: Perform a simple calculation**
   Although the overall process has multiple steps, each individual step involves basic operations (like multiplication, addition, and finding a square root). These simple calculations are the fundamental building blocks that lead to the final answer.
8. **Item 2: Select an appropriate strategy from a list**
   Students need to choose the correct mathematical method for the situation (e.g., using the Pythagorean theorem to find the slant height). Although the question doesn't provide a list of strategies, it implicitly requires the student to make this decision, involving a strategy selection process.
9. **Item 3: Implement a given strategy to determine a mathematical solution**
   Once a strategy is determined (e.g., "find the slant height using the theorem, then calculate the area"), the student must implement it by carrying out the calculations, putting the chosen plan into action.
10. **Item 13: Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy**
    The real-world context of a garage might hint at the reasonableness of the answer (e.g., units of measurement, expected size). However, as the question likely provides exact figures, this aspect is less critical than the others.
11. **Item 11: Switch between different representations in the process of finding solutions**
    The transition from the 2D plan diagrams to the actual 3D dimensions and calculations can be seen as switching between representations. However, this process is largely encompassed by Item 10, hence its lower ranking.

1. **第6条：从图表中提取数学信息**
   题目给出了两个平面图，学生必须从中读取必要的尺寸（如车库长度、宽度、屋顶高度等），这是解题的基础。正确提取信息是后续计算的前提，因此最为关键。
2. **第5条：理解并利用基于定义、规则和形式系统的构造，包括使用熟悉算法**
   计算屋顶面积需要运用矩形面积公式和勾股定理等几何规则，这些是学生熟悉的算法。通过应用这些规则，才能将尺寸转化为面积。
3. **第12条：使用多步骤程序得出数学解决方案**
   解题过程包含多个步骤：先根据宽度和高度求斜边长度，再计算单个矩形面积，最后乘以2得到总面积。这体现了多步推理和程序化思维。
4. **第8条：阐述解决方案，展示和/或总结中间结果**
   题目明确要求“Show your work”，即需写出计算过程，包括中间结果。这直接对应第8条，强调解题过程的呈现和沟通。
5. **第7条：操作数字、图形和统计数据、代数表达式和方程以及几何表示**
   计算中涉及数字运算（加减乘除、开平方）、代数表达式（勾股定理公式）以及几何量的转换，学生需要熟练操作这些数学对象。
6. **第10条：在解决问题时理解、关联和使用各种表示**
   平面图是一种图形表示，学生需理解其与实际尺寸的关系，并将图形信息转化为数值，这体现了对多种表示的理解和运用。
7. **第1条：进行简单计算**
   虽然整体步骤较多，但每一步都包含基础运算（如乘法、加法、平方根），这些简单计算是构成最终答案的基本单元。
8. **第2条：从列表中选择合适策略**
   学生需要根据问题情境选择恰当的数学方法（如用勾股定理求斜边），这涉及策略的选择，尽管题目未提供列表，但隐含了决策过程。
9. **第3条：实施给定策略以确定数学解决方案**
   在确定策略后，学生需执行计算，将策略付诸实践，这与第5条有重叠，但更侧重于执行环节。
10. **第13条：利用对上下文的理解指导或加速求解过程，例如以适合上下文的精度工作**
    车库的实际背景可能提示答案的合理性（如尺寸单位、精度），但通常题目已给出精确值，因此这条相对次要。
11. **第11条：在寻找解决方案的过程中在不同表示之间切换**
    从平面图到实际尺寸的转换可视为不同表示间的切换，但这一过程已包含在第10条中，故排名较后。